

DRUGS USED TO TREAT MIGRAINE

BY DR HAFSA SOHAIL

MIGRAINE

- Most common type of headaches are migraine, tension type and cluster headache.
- Patients with severe migraine headaches report 1 to 5 attacks per month of moderate to severe pain which is usually unilateral

DIFFERENCES

	MIGRAINE	CLUSTER	TENSION TYPE
Family history	Yes	No	Yes
Sex	Females more often than males	Males more often than females	Females more often than males
Onset	Variable	During sleep	Under stress
Location	Usually unilateral	Behind or around one eye	Bilateral in band around head
Character and severity	Pulsating, throbbing	Excruciating, sharp, steady	Dull, persistent, tightening
Duration	2-72 hours per episode	15-90 minutes per episode	30 minutes to 7 days per episode
Associated symptoms	Visual auras, sensitivity to light and sound, pale facial appearance, nausea and vomiting	Unilateral or bilateral sweating, facial flushing, nasal congestion, lacrimation, pupillary changes	Mild intolerance to light and noise, anorexia

MIGRAINE CHARACTERISTICS



- HEADACHE ATTACKS LASTING 4-72 HOURS
- UNILATERAL LOCATION
- PULSATING SENSATION
- MODERATE OR SEVERE PAIN INTENSITY
- AGGRAVATED BY ROUTINE PHYSICAL ACTIVITY
- ACCOMPANIED BY NAUSEA AND/OR VOMITING
- SENSITIVITY TO LIGHT AND/OR SOUND

TENSION-TYPE HEADACHE CHARACTERISTICS



- LASTING FROM 30 MINUTES TO 7 DAYS
- BILATERAL LOCATION
- PRESSING OR TIGHTENING SENSATION AROUND THE HEAD
- MILD OR MODERATE PAIN INTENSITY
- NOT AGGRAVATED BY ROUTINE PHYSICAL ACTIVITY
- NO NAUSEA OR VOMITING
- SENSITIVITY TO EITHER LIGHT OR SOUND

Migraine

- Positive family history
- Females more affected than males.
- Usually unilateral
- Pulsating, throbbing
- Visual aura, sensitivity to sound and light
- Nausea and vomiting

Types Of Migraine



More Than a Headache: The Many Faces of Migraine

01

Migraine Without Aura

Standard throbbing head pain, nausea, and light sensitivity.

02

Silent Migraine

All the symptoms of a migraine (nausea, aura) but no headache.

03

Menstrual Migraine

Attacks triggered by shifting hormones during a cycle.

04

Migraine With Aura

Visual "warning" signs (flashes/spots) followed by head pain.

05

Vestibular Migraine

Primarily causes vertigo (spinning) and balance loss.

06

Abdominal Migraine

Severe stomach pain and vomiting (mostly in children).

BIOLOGICAL BASIS OF MIGRAINE

- Pain of migraine may be due to intracranial and extracranial arterial vasodilation which leads to release of neuroactive molecules such as Substance P, neurokinin A, and calcitonin gene related peptide.
- With AURA Hypoperfusion
- Without AURA..... NO Hypoperfusion

DRUGS USED TO TREAT MIGRAINE

- Triptans (5-HT_{1b/1d})
- Ergot alkaloids (5-HT_{1d})
- Nonspecific therapies/ symptomatic treatment

NONSPECIFIC THERAPIES

- Analgesics
- Antiemetics
- Nonsteroidal inflammatory drugs.
- Corticosteroids

TRIPTANS

- Almotriptan
- Eletriptan
- Frovatriptan
- Naratriptan
- Rizatriptan
- Zolmitriptan

TRIPTAN

- Sumatriptan ---- first available triptan
- Prototype of triptan
- Rapidly reduce severity of migraine headaches in 70 % patients.
- Serotonin agonists acting found on small peripheral nerves that innervate intracranial vasculature.
- Sumatriptan is given subcutaneously, intranasally, or orally.

CONTINUE

- Zolmitriptan is available orally and by nasal spray.
- All other agents are taken orally .
- Frovatriptan is the longest triptan, with half life of more than 24 hours.
- Elevation of blood pressure is associated with triptan use.

ADVERSE EFFECTS

- PRESSURE SENSATION:
- Chest
- Neck
- Throat
- Jaw
- DIZZINESS, MALAISE

ERGOT ALKALOIDS

- Ergotamine and dihydroergotamine
- Bind to 5-HT₁ receptors , alpha receptors, and dopamine receptors.
- Available sublingually.
- Ergotamine is used with strict daily and weekly dosage limits due to its availability to cause dependence, Rebound headache
- Dihydroergometrine is administered intravenously or intranasally.

MOA & ADVERSE EFFECTS

- Targets 5HT-1 receptor in intracranial blood vessel and cause vasoconstriction
- ADR
- NAUSEA
- CONTRAINDICATED IN ANGINA AND PERIPHERAL VASCULAR DISEASES

Prophylaxis

- DRUG OF CHOICE B-BLOCKER
- EG PROPRANOLOL, METOPROLOL

ALTERNATIVES

CA-CHANNEL BLOCKER

ANTICONVULSANTS

ANTIDEPRESSANT

Agents used to treat an acute attack

TRIPTANS

- Triptans rapidly and effectively abort or markedly reduce the severity of migraine headaches in about 80% of patients.
- Triptans are serotonin agonists, acting at 5-HT_{1D} receptors.

DIHYDRO- ERGOTAMINE

- Dihydroergotamine is a vasoconstrictor.
- Most effective when given during the prodromal phase.
- Contraindicated in pregnancy and in patients with peripheral vascular disease or coronary artery disease.

ANALGESICS

- Anti-inflammatory drugs, such as *aspirin*, *naproxen*, and *ibuprofen*, are useful in relieving migraine attacks.
- Severe pain may require administration of opioids, such as *codeine sulfate* or *meperidine*.

Agents used in prophylaxis

Several classes of drugs are effective in reducing the frequency and severity of migraine attacks:

- β -Blockers: *Propranolol* and *nadolol*.
- Tricyclic antidepressant: *Amitriptyline*
- Anticonvulsant: *Divalproex*
- Calcium channel blocker: *Verapamil*

Time

Start of
attack

Asymptomatic phase

- Between attacks, no symptoms or pathologic features are evident.

Prodromal phase

- Visual disturbances that precede the actual headache.
- Associated with arterial vasoconstriction and release of serotonin.

Headache phase

- Pain, nausea and vomiting.
- Associated with cerebral vasodilation and lower-than-normal levels of serotonin.

THANK YOU